

```
#include <Geotopia>
#include <Kernel File>
#include <string>
int main() {
    std::ifstream inputFile("vocabulary.txt"); // Open a geometry file named
vocabulary.txt
```

Internet Security of Geotopia



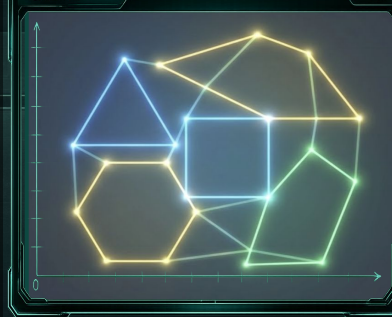
Access Granted

DEFINITION

Regular Polygon

A polygon that has all equal side lengths and all equal angle measures. If the sides or angles of a polygon are not equal, the polygon is irregular.

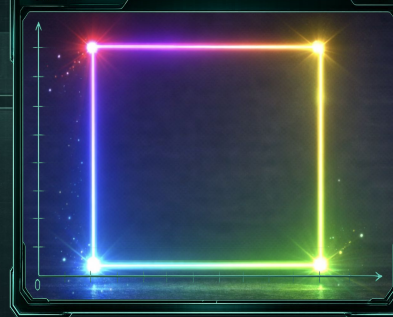
GRAPH



DEFINITION

Square is a regular polygon with 4 sides.

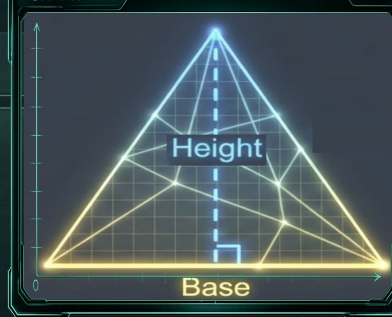
GRAPH



DEFINITION

triangle is a polygon with 3 sides.

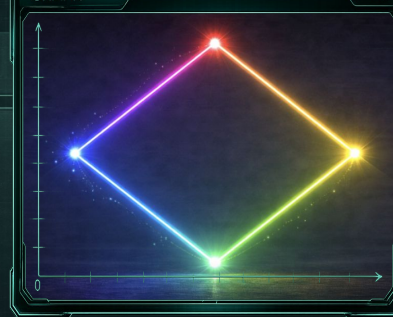
GRAPH



DEFINITION

Rhombus has four equal side lengths. A rhombus does not necessarily have four equal angles.

GRAPH



DEFINITION

An isosceles triangle has 2 equal sides and angles.

An equilateral triangle has 3 equal sides and angles.

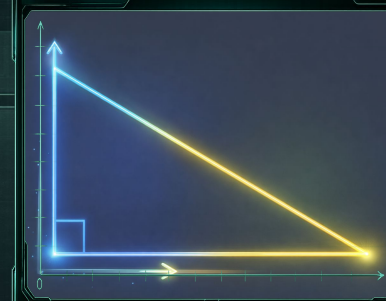
GRAPH



DEFINITION

A right triangle has one right angle. Right angles measure exactly 90° .

GRAPH



DEFINITION

An acute triangle has three acute angles. Acute angles measure less than 90° .

GRAPH



DEFINITION

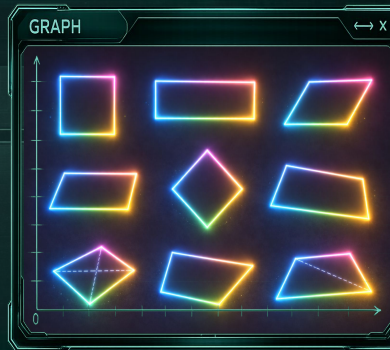
An obtuse triangle has one obtuse angle. Obtuse angles measure more than 90° .

GRAPH



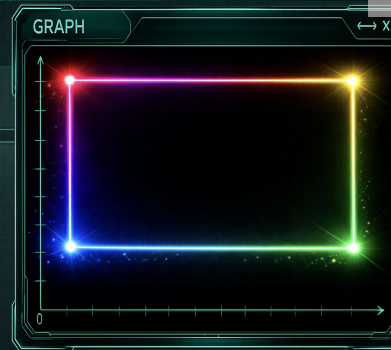
DEFINITION

Quadrilateral is a two-dimensional shape with four sides and four angles.



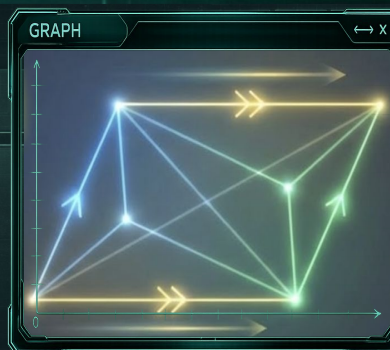
DEFINITION

Rectangle is a parallelogram with four right angles.



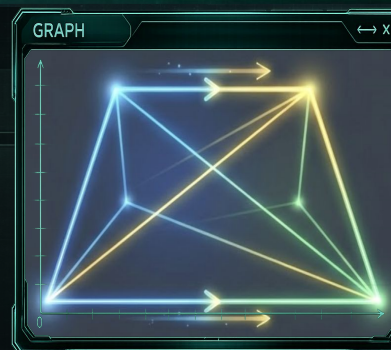
DEFINITION

Parallelogram is a quadrilateral with two pairs of parallel sides.



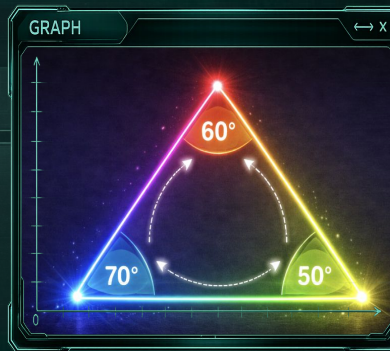
DEFINITION

Trapezoid is a quadrilateral with exactly one pair of parallel sides. To find the area, multiply height by half of the two bases.



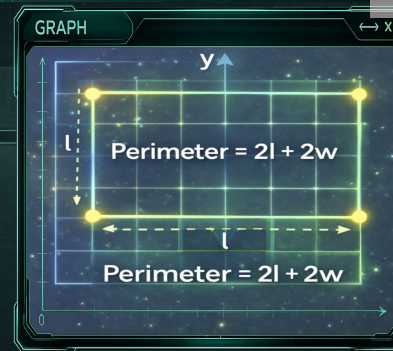
DEFINITION

Triangle Angle-Sum: The three interior angles in any triangle always add up to 180° .



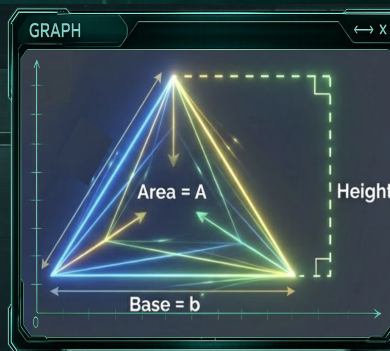
DEFINITION

Perimeter is the distance around the outside of a two-dimensional shape.



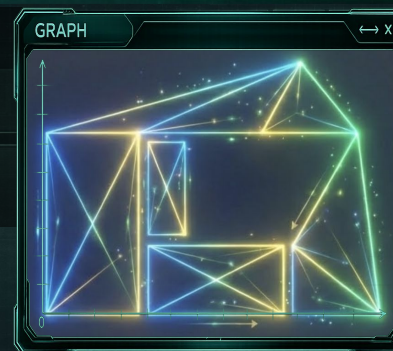
DEFINITION

Area is the space that a polygon takes.



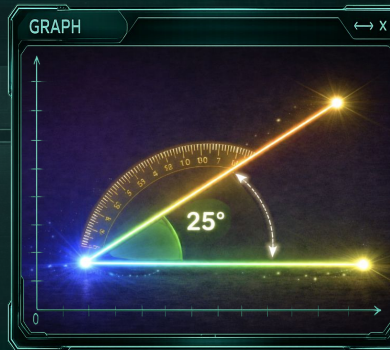
DEFINITION

A composite figure is a polygon made up of basic shapes put together.



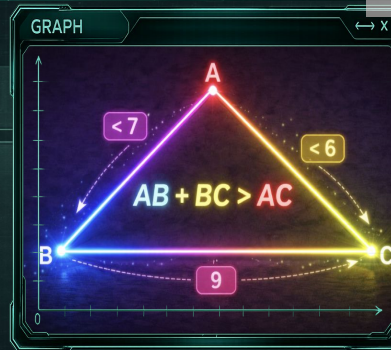
DEFINITION

Angle measure is between 0 degrees (written as 0°) and 360° . Protractor is a tool used to measure angles.



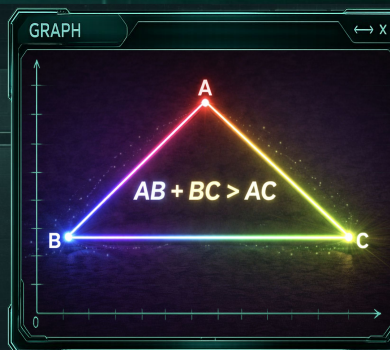
DEFINITION

Triangle inequality: The sum of any two sides of a triangle must be greater than the third side. The difference between any two sides must be less than the third side.



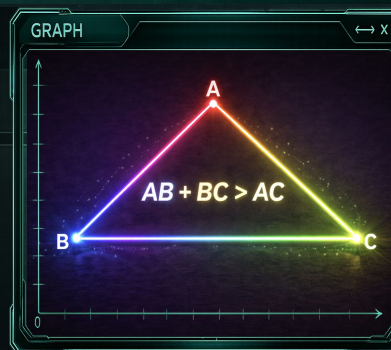
DEFINITION

Possible lengths of a triangle are between the sum and the difference of the other two sides.



DEFINITION

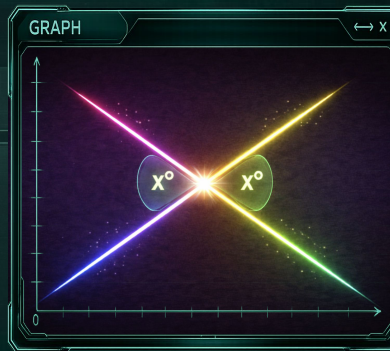
Example: If a triangle has sides of 2 and 6, then the length of the third sides are between $(6-2=4)$ and $(6+2=8)$ which are 5, 6, 7.



DEFINITION

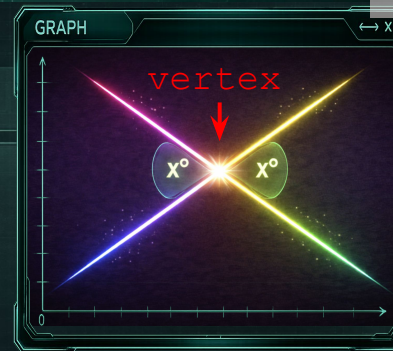
Vertical angles

Two Angles are supplementary when they add up to 180 degrees.



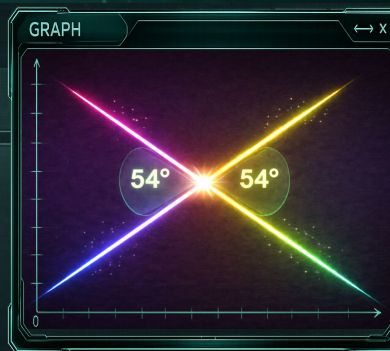
DEFINITION

Vertical in geometry means they share the same vertex (corner point).



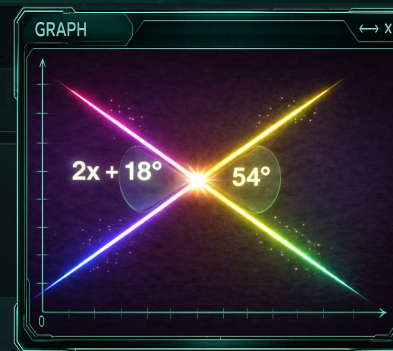
DEFINITION

If two angles are vertical, then they must be in equal measure.



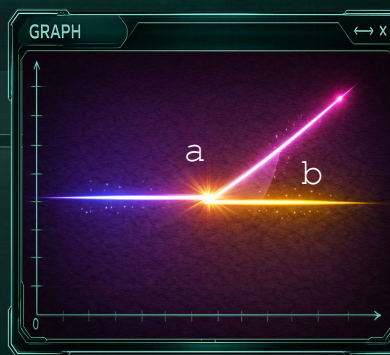
DEFINITION

Example: Angle A and Angle B are vertical angles. Then we can write out $2x+18=54$ and solve for x .



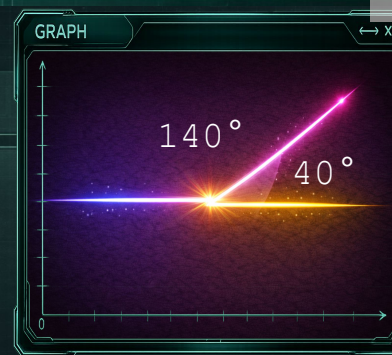
DEFINITION

Supplementary angles: are the angles opposite to each other when two lines cross. They share same angle measure.



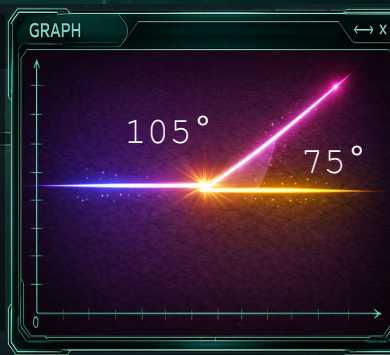
DEFINITION

If two angles are supplementary, then they must add up to 180 degrees.



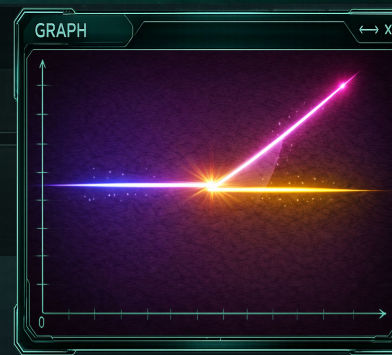
DEFINITION

Example: angles with 140 degree and 40 degrees are supplementary. Angles with 75 degree and 105 degrees are supplementary.



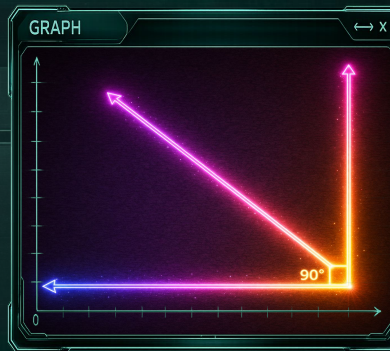
DEFINITION

Supplementary angles usually appear on one side of a line and share the same side.



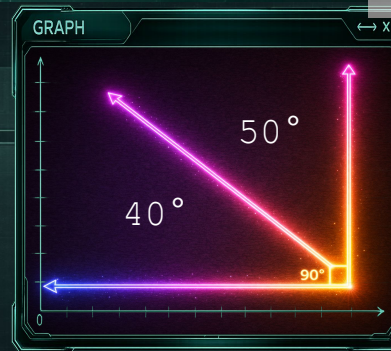
DEFINITION

Complementary angles: Two Angles are complementary when they add up to 90 degrees (right angle).



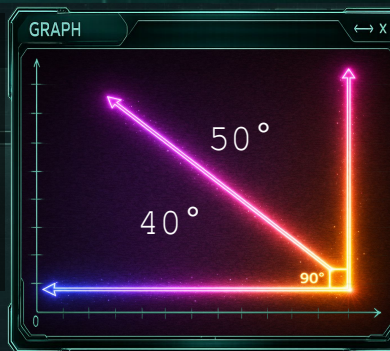
DEFINITION

If two angles are complementary, then they must add up to 90 degrees.



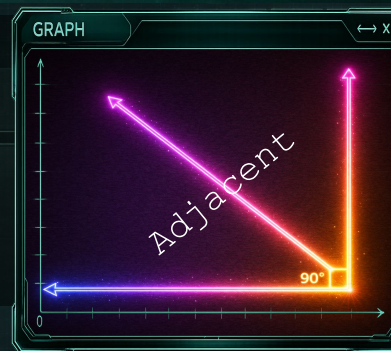
DEFINITION

Example: angles with 50° and 40° are complementary; angles with 75° and 15° are complementary.



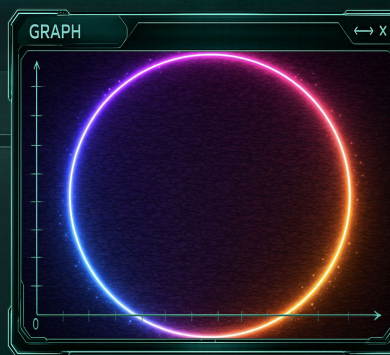
DEFINITION

Adjacent angles: If two angles are side by side and share the same vertex, they are adjacent angles.



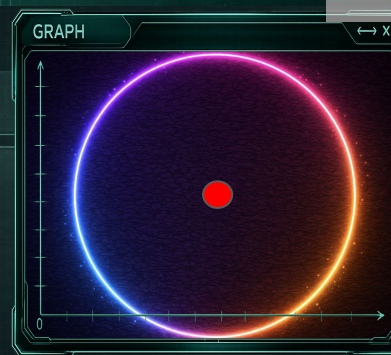
DEFINITION

Circle is a round shape made by all the points that are the same distance from a center.



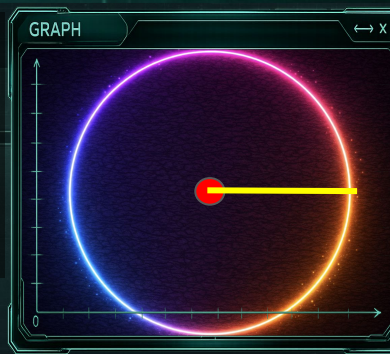
DEFINITION

Center: the central point of the circle.



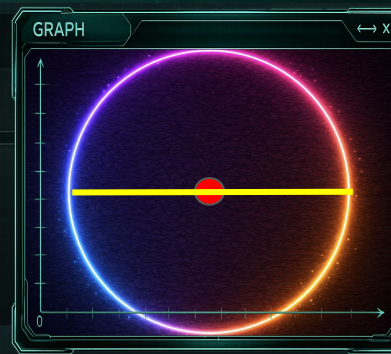
DEFINITION

Radius: The distance from the center to any point on the circle.



DEFINITION

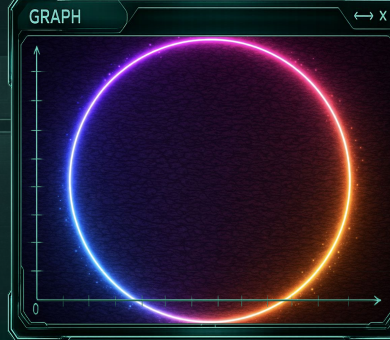
Diameter: A line that passes through the center of the circle and has its endpoints on the circle. The diameter is twice the length of the radius.



DEFINITION

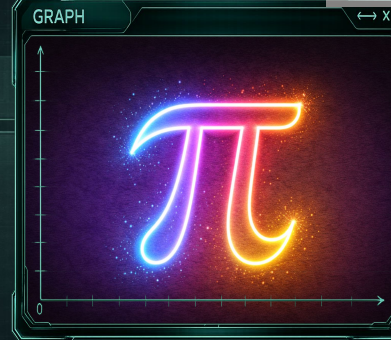
Circumference:

The distance around the circle. It's the circle's perimeter.



DEFINITION

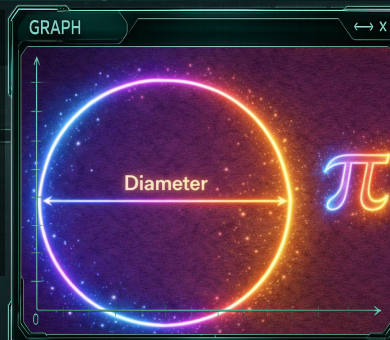
π (pi): a special number (approximately 3.14) that represents the ratio of circumference to diameter.



DEFINITION

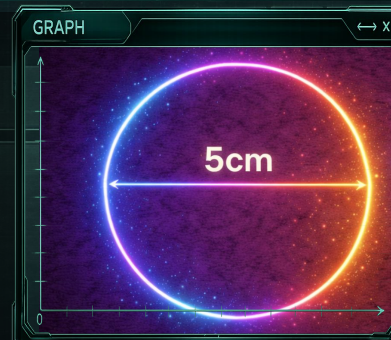
How to find circumference:

It's always π times the diameter.



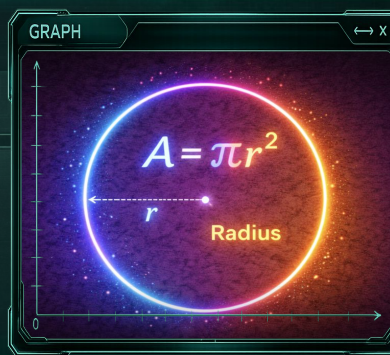
DEFINITION

Example: If diameter is 5cm then circumference is 5π cm (in terms of π) or $5 \times 3.14 = 15.70$ cm (in decimal).



DEFINITION

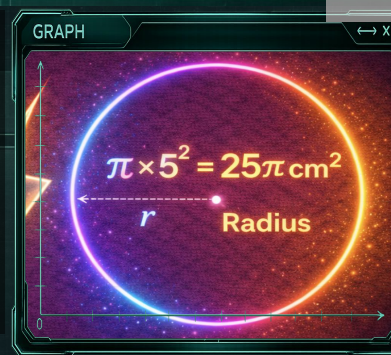
Area: is the amount of space inside the circle which is multiplying π or 3.14 by the square of the radius.



DEFINITION

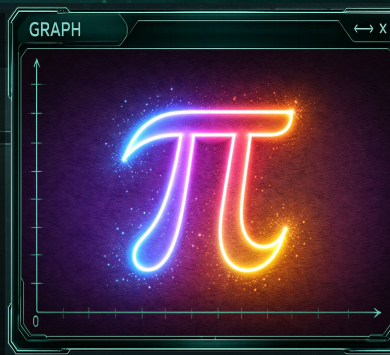
How to find area:

If radius is 5cm then area is $\pi (5^2) = 25\pi \text{ cm}^2$ (in terms of pi) or $25 \times 3.14 = 78.50 \text{ cm}^2$ (in decimal).



DEFINITION

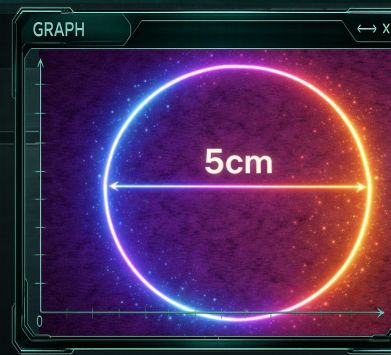
From circumference to area: If the circumference is 8π , then the diameter is $8\pi/\pi = 8$, radius is $8/2 = 4$, and area is $\pi (4^2) = 16\pi$ or 50.24.



DEFINITION

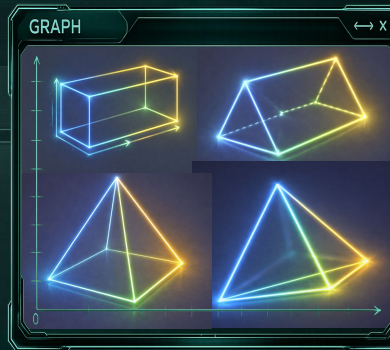
Another Example:

If the diameter is 5cm, then the radius is $5/2 = 2.5 \text{ cm}$, area is $\pi (2.5^2) = 6.25\pi \text{ cm}^2$ or $6.25 \times 3.14 = 19.63 \text{ cm}^2$.



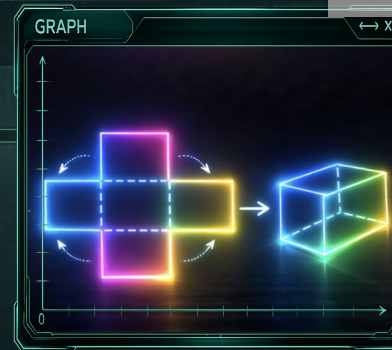
DEFINITION

Polyhedron (Polyhedra) is a 3D shape with faces, edges, and vertices. Examples include prisms, cubes and pyramids.



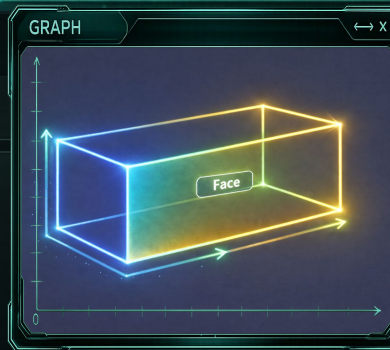
DEFINITION

Net: a flat pattern that, when folded, makes a 3D shape. Net is used to calculate the surface area.



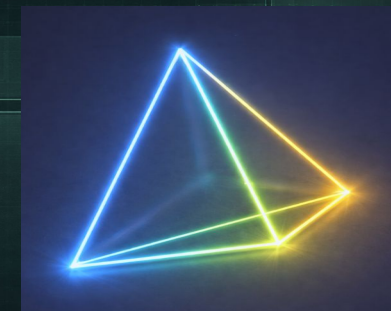
DEFINITION

A **prism** is a 3D shape which two opposite faces must be congruent (identical in size). For example, both top and bottom are rectangles or triangles.



DEFINITION

A **pyramid** is a 3D shape with a polygon base and a pointed top.



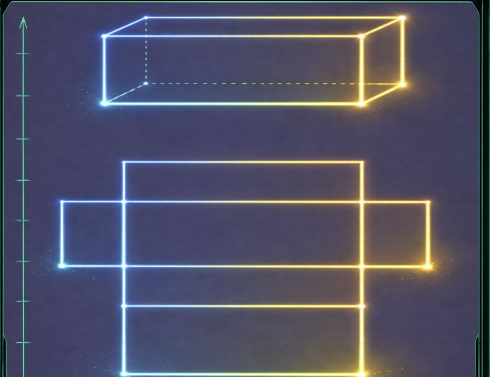
DEFINITION



Surface area: The sum of the area of all faces of a 3D shape.

How to find surface area: find the area of each face, then add the answers up.

GRAPH



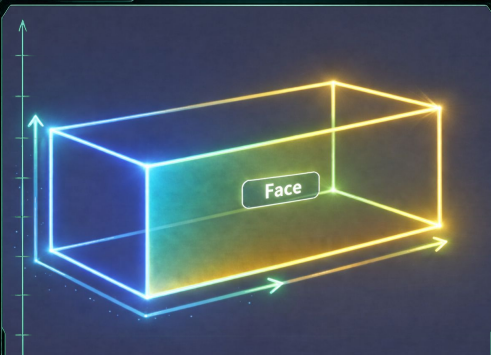
DEFINITION



Surface area of a Rectangular

Prism: Simply find the product (multiplication) of **length** and **width**, then the product of **length** and **height**, then the product of **width** and **height**. Add them up and double it.
In formula: $SA = 2(lw + lh + wh)$

GRAPH

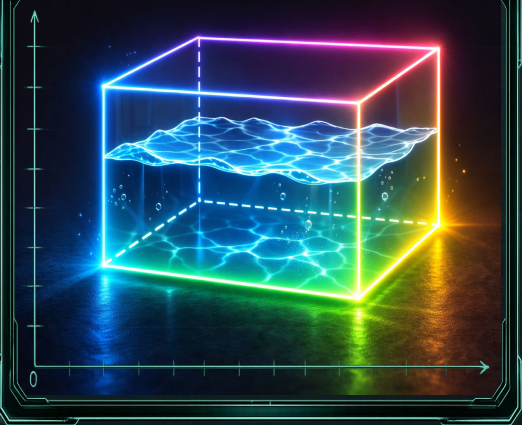


DEFINITION



Volume of a Rectangular Prism:
is length times width times
height. In formula: $V=l*w*h$

GRAPH



DEFINITION

Volume of any Prism is the
base(bottom) area times the
height.

If it is a triangular prism, the
volume is the base triangle
times the height.

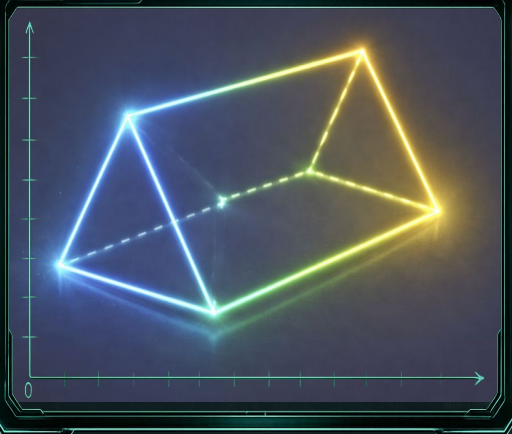
If it is a rectangular prism,
the volume is the base rectangle
times the height.

If it is a cylinder, the volume
is the base circle times the
height.

DEFINITION

Volume of a triangular prism: is the base **triangle** area times the **length** (which is the height if we "stand" the prism up). The formula is $V = (\frac{1}{2} * b * h) * l$.

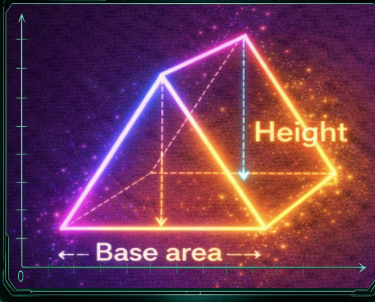
GRAPH



DEFINITION

Surface area of a triangular prism: is the sum of its 5 faces. The front and back are both triangles. They are congruent (same area). The bottom is a rectangle (base*length). The left and right are rectangles (slant height*length). The formula is very long. We should remember the net and ways to find area of each face.

GRAPH

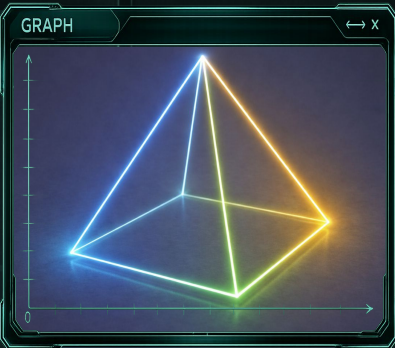


DEFINITION

Volume of any pyramid is $\frac{1}{3}$ the base area times its height. This means "triple the volume of a pyramid is volume of a prism".

Volume of square pyramid is $\frac{1}{3}$ the base square (base*base) times its height.

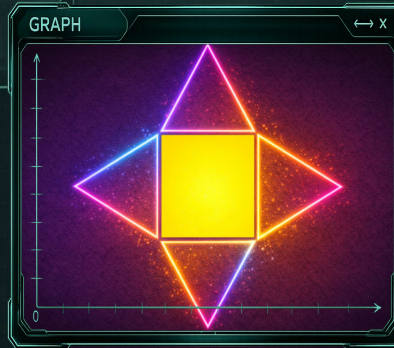
GRAPH



DEFINITION

Surface area of square pyramid is the sum of 5 faces. Look at the net of it. We see that the 4 triangles around are congruent (same area). The surface area is 4 of the triangle area (slant height*base) times the height.

GRAPH



DEFINITION



Cross section: Imagine cutting a solid in any direction. The shape we get when cutting is the cross section.

GRAPH



DEFINITION

Sides of a cross section:

When you cut the prism or pyramid, the sides **parallel** and **perpendicular** (right angle) to your "knife" are the side of your cross sections.

For example, when cutting a rectangular prism horizontally, the sides of the cross section are **the length** and **the width**.